Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for:
Cardiac Disease in Pregnancy
Eastern and Northern Maternity Services

Summary

This guideline outlines the process of management and referral for pregnant women with cardiac disease.

Key Points

The essential elements of this guideline are:

1. Assess cardiac status early
2. Refer to cardiologists early
3. Refer to anaesthetics early
4. Discuss with Bristol early
5. Refer to Regional Maternal Medicine Network weekly cardiac MDT (see appendix 5)
6. Treat any deterioration in maternal condition seriously

Refer to the 2018 European Society of Cardiology Guidelines for the Management for Cardiovascular Diseases during Pregnancy for further information and specific management.

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1. INTRODUCTION

1.1 Cardiac disease was the leading cause of maternal deaths in the period 2019 – 2021 (MBRRACE 2022)

1.2 Cardiovascular disease in pregnancy has increased due to increasing age at first pregnancy and increasing prevalence of diabetes, hypertension and obesity.

1.3 An increasing number of women with congenital heart disease are reaching childbearing age.

2. PHYSIOLOGICAL ADAPTATIONS TO PREGNANCY

2.1 Plasma volume and cardiac output rise by 40-50% above baseline by 32 weeks of gestation and most of this occurs in the first trimester.

2.2 Atrial and ventricular diameters increase but in women with heart disease, the ventricular adaptation may be suboptimal.

2.3 Cardiac dysfunction is related to suboptimal uteroplacental flow and poor fetal outcomes.

2.4 Blood pressure and cardiac output increase during labour and postpartum.

2.5 Supine positioning, pain, anxiety, exertion, haemorrhage, anaesthesia and infection may all add additional stress on the cardiovascular system

2.6 Interventions and treatments used in an obstetric setting (IV fluids, oxytocin) may aggravate heart conditions and care must be planned accordingly.

3. PRE-CONCEPTUAL COUNSELLING

3.1 Pre-pregnancy risk assessment and counselling is indicated in all women with known or suspected congenital or acquired cardiovascular and aortic disease or pulmonary hypertension.

3.2 Informed maternal decision-making and individualised care is crucial, taking into account the medical condition, emotional/ cultural context and psychological issues. It may be carried out by local obstetricians & cardiologists and/or if complex by the tertiary centre (Bristol).

3.3 Points to cover include:

- Long term prognosis.
- Fertility and miscarriage rates.
- Estimated maternal risk and outcome.
- Women with cardiac disease have an increased risk of obstetric complications, including premature labour, pre-eclampsia, and postpartum haemorrhage.
- Expected fetal outcome.
- Offspring complications occur in 18-30% chance of patients with heart disease (1-4% neonatal mortality).
- Plans for pregnancy care and delivery.
- Recommendations for healthy lifestyle, weight and stopping smoking if relevant.

- Cardiac function should be assessed prior to pregnancy. As a minimum an ECG, ECHO and an exercise stress test should be performed. A pregnancy exercise capacity >80% is associated with a favourable pregnancy outcome.
- Modifying medication may be required.
- Contraceptive options should be discussed.

3.4 Patients with **pulmonary hypertension**, (also Eisenmenger's syndrome and vascular Ehlers-Danlos syndrome) should be advised **not** to get pregnant as they have a significant risk of mortality. Termination of pregnancy is justified. Otherwise care should be managed in Bristol.

4. CONTRACEPTIVE CHOICES

4.1 Progesterone-only contraception is likely to be safe for most women with a cardiac problem including the Mirena, progesterone only pill, Depo-Provera.

4.2 The copper IUCD is also suitable but may have a slightly increased risk of Introducing infection.

4.3 Combined oral contraception is safe if there is no increased risk of thrombosis with the heart condition.

4.4 Hormonal methods may be ineffective if the woman is using Bosentan (adrenergic inhibitor) for pulmonary hypertension.

5. PREGNANCY MANAGEMENT

5.1 **Pre-existing known cardiac conditions** should be managed according to the diagnosis. The ESC guidelines and Bristol Tertiary Unit Care pathway (see appendix 1) are useful to make individual management plans.

5.2 All pregnant women with cardiac conditions should be referred for an anaesthetic opinion. They should be referred to the Consultant Anaesthetic Antenatal clinic (select **Obstetric** not general high-risk anaesthetic clinic on MyCare system) and offered an appointment.

5.3 Close co-operation between the anaesthetist, cardiologist and obstetrician is important to ensure optimal management. Details of women with particularly high risk of complications from cardiac disease in pregnancy should be shared between obstetricians, obstetric anaesthetists and matrons on LW and Triage to ensure all relevant parties are aware of risk factors and care plan.

5.4 Every case has to be managed on an individual basis but general principles apply. Care should be supervised by an experienced obstetrician and a cardiologist. Some women may be advised to deliver in a tertiary centre.

5.5 **Cardiac disease may present in pregnancy.** A thorough booking history asking targeted questions may identify a problem that the woman herself may have not deemed significant

5.6 Risk factors for cardiovascular disease include:

- hypertension
- diabetes
- obesity (BMI >30)
- family history of heart disease
- aged over 35 years old
- smoker
- specific immigrant populations

5.7 Record pre-pregnancy symptoms, (if any), for example using the New York Heart association classification (see appendix 4).

5.8 Take a careful family history to include congenital heart disease and early (<60years) morbidity from IHD. Note history of sudden death in family members.

5.9 Cardiac disease should be considered in the following:

- Unexplained tachycardia
- Hypotension
- New onset shortness of breath
- Oxygen saturations < 94% or fall to < 94% on exertion
- Chest pain
- Palpitations associated with symptoms such as collapse or syncope
- Syncope during exercise can suggest a cardiac origin

5.10 Perform an ECG. Consider ABG, ECHO and CXR plus cardiology referral. May need a 24 hr holter monitor tape.

5.11 Chest pain ‘red flags’:

- Pain requiring opioids
- Pain radiating to arm, shoulder, back or jaw
- Sudden onset, tearing or exertional chest pain
- Associated with haemoptysis, breathlessness, syncope or abnormal neurology.

5.12 Perform an ECG. Consider ABG, ECHO and CXR plus urgent cardiology referral.

5.13 Think heart failure if:

- Orthopnoea (ruling out aortocaval compression) or at rest
- Unexplained cough, particularly when lying down or which produces frothy pink sputum
- Paroxysmal nocturnal dyspnoea – being woken from sleep by severe breathlessness and coughing, which may produce pink frothy sputum and is improved by moving to an upright position
- Associated palpitations

5.14 Perform ECG, request an ECHO and urgent cardiology consult. Consider an ABG and CXR.

6. IF CARE IS ADVISED IN THE LOCAL UNIT RATHER THAN TERTIARY CENTRE

6.1 Use the tertiary referral centre advice to help guide treatment. Unless otherwise stated in their recommendations:

6.2 Arrange to see at least every 4 weeks to monitor maternal symptoms and look for signs of deterioration in cardiac function, aiming to prevent complications.

6.3 Assessment during pregnancy should include:

- Blood pressure, pulse and rhythm every assessment.
- Listening to lung bases to detect pulmonary oedema
- Oxygen saturation every trimester.

6.4 If any deterioration then perform:

- electrocardiography (ECG)
- echocardiography
- blood gases

6.5 Make sure there is a plan for delivery and place of delivery made by 32-34 weeks. Labour care should take place in a unit with adequate facilities for cardiac monitoring if required and full resuscitation facilities.

6.6 In the majority of cases, the onset of labour should be spontaneous with vaginal delivery unless there is an obstetric reason for a caesarean. Exceptions would be patients presenting in labour on oral anticoagulants, with aggressive aortic pathology, in acute intractable heart failure or with severe pulmonary hypertension.

6.7 Fluid balance is paramount. Intravenous fluids should be used cautiously as the increased circulating blood volume may cause pulmonary odema. During active labour the left lateral position or upright position is advantageous to reduce aortocaval compression.

6.8 Timing of induction will depend on cardiac status, obstetric evaluation including cervical assessment, fetal well-being and fetal lung maturity

6.9 Induction of labour should be considered at 40 weeks gestation in all women with cardiac disease to reduce the risk of emergency caesarean section.

6.10 Both misoprostil (25mg, prostaglandin E1 (PGE1) or dinoprostone (1-3mg slow release formulation of 10mg (PGE2) can be safely used to induce labour.

6.11 Mechanical methods such as a cervical ripening balloon might be preferable in patients where a drop in systemic vascular resistance would be detrimental. Artificial rupture of membranes and infusion of oxytocin can be used safely in women with heart disease.

7. LABOUR

7.1 On admission assess cardiac status, review plan for delivery and inform senior obstetric and anaesthetic staff. Epidural is the best form of pain relief. Care must be taken with oxytocin use for augmentation as may can cause fluid retention, tachycardia and reduce afterload. Oxytocics should be discussed with the consultant on call.

7.2 Pain during labour may cause sympathetic response increasing heart rate and blood pressure. Strongly consider a carefully titrated epidural.

7.3 Midwives must maintain a strict fluid balance throughout labour and the initial postnatal period. Blood pressure and heart rate should be carefully monitored,

7.4 Management of the second stage

7.4.1 The active phase of the second stage should be delayed for 2 hours to allow maximal descent of the fetal head.

7.4.2 Pushing in the second stage may reduce the venous return. It should be kept short and without exertion – therefore prolonged active pushing is not advocated. It may be considered optimal to perform an assisted delivery with vacuum extraction or forceps where appropriate or possible.

7.5 Management of the third stage

7.5.1 Low dose Oxytocin infusion can usually be used safely but if there is significant postpartum haemorrhage mechanical methods such as uterine compression and or brace sutures should be employed early. Women with impaired cardiac reserve tolerate PPH poorly.

8. POSTNATAL PERIOD

8.1 The risk of cardiac failure with pulmonary oedema is greatest in the early puerperium.

8.2 Signs and symptoms include:

- tachycardia
- cyanosis
- oedema
- distension of the liver

8.3 If pulmonary oedema occurs acute dyspnoea with frothy sputum and haemoptysis may occur. The woman will require High Dependency Care possibly ITU to stabilise condition.

8.4 A critical period is 4-5 days post-delivery due to the risk of thrombus formation and pulmonary embolism (recommend at least 10 days thromboprophylaxis).

8.5 Postnatal follow up in a multidisciplinary clinic should be considered to discuss relevant issues and plans for further pregnancies.

8.6 Meticulous leg care, elastic support stockings, and early ambulation are important to reduce the risk of thrombo-embolism.

8.7 Breastfeeding should be encouraged unless specifically contraindicated in the presence of certain medications. Expert opinion on the safety of any maternal medications in relation to pregnancy should be sought, ideally prior to delivery.

9. MONITORING COMPLIANCE WITH THIS GUIDELINE

9.1 Any concern or non-compliance with this guideline that is identified through the investigation of clinical incidents, claims or complaints will be reviewed as per the Trust Policies regarding Incidents, Claims and Complaints, and may result in an audit and/or amendment to the guideline.

10. ASSOCIATED CLINICAL GUIDELINES OR POLICIES

- VTE Guideline

11. PUBLICATION DETAILS

Author of Clinical Guideline	Jen Blackman
Division/ Department responsible for Clinical Guideline	Specialist Services/CWH/Maternity
Contact details	jennifer.blackman@nhs.net
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APPENDIX 1:

BRISTOL REGIONAL PATHWAY

Cardiomyopathies

Previous peripartum cardiomyopathy Dilated / Hypertrophic cardiomyopathy	Initial assessment will usually take place at the local centre. Discussion with the regional centre is encouraged. Place of further antenatal care and delivery will be dependent upon the severity and local capacity for management. Multidisciplinary plan will be available to the local centre.

Ischaemic Heart Disease

Acute coronary syndrome	Treat as per usual ACS protocol for non-pregnant women (ie. PCI) ESC guidance should be used in decision making Discussion with the regional centre may be appropriate. Plan of care should be made jointly with local obstetricians, anaesthetists and cardiologists.
Stable ischaemic heart disease	Discussion with the regional centre is appropriate.

BRISTOL REGIONAL PATHWAY

Arrhythmias / Pacemakers

- Initial cardiology assessment at the local centre (see ESC guidelines 2011).
- Discussion with the regional centre may be appropriate.
- Plan of care to be made jointly with local obstetricians, anaesthetists and cardiologists.
- Potential to deliver at local centre dependent on perceived risk and facilities available.

Acquired Valvular Heart Disease

Mechanical heart valves	✦ Assessment, preferably within 6 weeks of conception, by regional centre if possible. ✦ Anticoagulation strategy available for counselling purposes on website ✦ Discussion with the regional centre for further antenatal care and delivery, dependent upon severity and/ or local capacity.
Acquired mitral and aortic valve disease, including rheumatic heart disease and bioprosthetic heart valves	✦ Initial assessment by cardiologist at the local centre. ✦ Place of further antenatal care and delivery dependent upon severity of valvular lesion (see Congenital Cardiac Disease pathway). ✦ All moderate aortic and mitral stenosis is potentially high risk and should be referred to the regional centre for assessment

APPENDIX 2 - CONGENITAL HEART DISEASE

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Low risk	✦ Patent Ductus Arteriosus (isolated without pulmonary hypertension) ✦ Moderate mitral/aortic regurgitation ✦ Mild aortic stenosis ✦ Total anomalous pulmonary venous drainage repair ✦ Bicuspid aortic valve, no aortopathy ✦ Repaired ASD/VSD ✦ Mild/moderate pulmonary stenosis	✦ Assessment by consultant cardiologist in local centre who may wish to discuss with ACHD cardiologist
Moderate risk	✦ Repaired coarctation with no sequelae ✦ Repaired Fallot's ✦ Severe pulmonary stenosis/ regurgitation ✦ Bicuspid aortic valve aortopathy < 45mm ✦ Mild mitral stenosis ✦ Severe aortic/ mitral regurgitation ✦ Ebstein's anomaly	✦ Initial assessment by consultant cardiologist in local centre. Referral or discussion with ACHD cardiologist is recommended. ✦ Likely to require 1-2 visits to the regional centre, but majority of antenatal and delivery can take place locally. ✦ Multidisciplinary plan/ advice will be available in the maternity notes if seen at the regional centre.
High risk	✦ Transposition of the great arteries: following Mustard or Senning repair or congenitally corrected transposition ✦ Transposition of the great arteries: arterial switch repair ✦ Moderate mitral stenosis ✦ Moderate/severe aortic stenosis ✦ Fontan circulation ✦ Cyanotic heart disease (without pulmonary hypertension) ✦ Native coarctation/ re-coarctation ✦ Marfan's and other serious vascular connective tissue disorders or Turner's syndrome with maximum aortic root/ Asc Aorta <45mm ✦ Bicuspid aortic valve aortopathy 45-50mm	✦ Referral to the regional centre is advised ✦ Specialist antenatal care and delivery will usually be at the regional centre, but normal antenatal care and some secondary cardiac care may take place locally ✦ A multidisciplinary plan will be available to the local centre should an unexpected local admission occur.
Very High Risk	✦ Pulmonary arterial hypertension (including Eisenmenger syndrome) ✦ Marfan Syndrome (or other vascular CTD*/ Turners syndrome) with aortic root/ Asc Aorta dilatation >45mm or ✦ Bicuspid aortic valve > 50mm ✦ Severe mitral stenosis	✦ Specialist antenatal and full cardiac care and delivery at the regional centre recommended.

*Includes Ehlers Danlos Type 4, Loeys-Dietz, Familial Thoracic Aortic Aneurysm and Dissection Syndrome.

APPENDIX 3: BRISTOL CONTACT DETAILS

Regional Centre: University Hospitals Bristol Cardiac Obstetric Clinic Fax 0117 3425688 cardiologist and local cardiology team if previously seen in Bristol or another centre. Inform them of the pregnancy and ask for advice.

CARDIOLOGISTS	OBSTETRICIANS	ANAESTHETISTS
Dr Stephanie Curtis	Dr Johanna Trinder	Dr Mike Kinsella
Dr Yasmin Ismail	Dr Aarthi Mohan	Dr Mark Scrutton
Dr Graham Stuart	Dr Louise Ashelby	Dr Issie Gardner
0117 3425967 / 6576	0117 3435250	Dr Neil Muchatua
Dr Nick Wharton		
Dr Clare Dowse		
0117 342 520		

APPENDIX 4: NEW YORK HEART ASSOCIATION CLASSIFICATION OF SYMPTOMS

NYHA	
I	No symptoms or limitation of ordinary physical activity
II	Mild symptoms, e.g. short of breath and/or angina and slight limitation during ordinary activity
III	Marked symptoms even on walking short distances, (10-20m) only comfortable at rest
IV	Severe limitations, symptoms even at rest, may be bed-bound

APPENDIX 5: Regional Maternal Medicine Network - Cardiac MDT

Regional Maternal Medicine Network has a weekly cardiac MDT.

To refer a patient for discussion at this MDT (preconception advice or in pregnancy) go to referapatient.org and then select "refer a patient" from the selection of options top right. Then type in "South West Maternal Medicine Network, North Bristol NHS Trust" followed by "Maternal Medicine Centre" in the box below this. Enter as many relevant clinical details as possible. You can ask to join the online meeting to discuss the case if you are able to. A response will come via email. Please make sure the referral and response are uploaded into the media section of Epic and a note written that the MDT discussion has taken place.